Determinants of Happiness in the United States, 2017

Group members:

Abdelrahman Ahmed Youssef 5220913

Menna-Tallah Mostafa Elsayed 5220819

Mona Ahmed Abdelmonem 5220205

Salma Khaled Ahmed 5220475

Reham Mohamed El-demerdash 5220671

Under supervision:

Dr/Laila El-Zeini

Table of contacts:

Introduction:

Happiness, or subjective well-being, has become a central topic in social sciences as researchers seek to understand the factors that shape individuals' quality of life. Measuring happiness allows policymakers, psychologists, and economists to better understand how social, economic, and cultural conditions influence people's lived experiences. Instead of relying only on traditional economic indicators such as income or employment, modern research increasingly emphasizes personal perceptions, values, and life circumstances.

The United States presents a unique context for studying happiness due to its cultural diversity, economic variation, and wide range of social values. The World Values Survey (WVS) offers an important opportunity to examine these dynamics using nationally representative data. In 2017, the WVS collected detailed information from U.S. respondents on their beliefs, financial satisfaction, demographic characteristics, and subjective well-being.

Happiness is used as the dependent variable, while a set of social, economic, and demographic indicators—including importance of religion, tolerance toward others, respect for authority, financial satisfaction, sex, age, urban versus rural residence, and educational level are examined as potential predictors.

By exploring how these factors are associated with happiness, this research provides insights into the social and economic foundations of subjective well-being in the United States. The findings can help inform public policy, contribute to academic literature, and expand understanding of how personal values and life circumstances shape happiness in modern American society.

The Primary Aim of the Project:

To analyze the factors associated with happiness in the United States in 2017, and to examine the relative explanatory power of financial satisfaction, core social values, and standard demographic characteristics as independent variables

Variables Used in the research:

Variable	WVS Code	Classification	Concept Measured
Happiness (Dependent Variable)	**Q46**	Subjective Well-being	The respondent's self-reported level of happiness.
Importance of Religion	**Q6**	Social Values	How important religion is in the respondent's life.
Value of Tolerance	**Q12**	Social Values	How important tolerance and respect are as a quality children should be taught.
Respect for Authority	**Q45**	Social Values	Attitude towards the necessity of showing greater respect for authority.
Financial Satisfaction	**Q50**	Economic Factors	Satisfaction with the household's financial situation.
Sex/Gender	**Q260**	Demographic	The respondent's sex (Male/Female).
Age Group	**X003R2**	Demographic	The respondent's age (recoded into age categories).
Settlement Type	**H_URBRURAL**	Demographic	Type of residence area (Urban vs. Rural).
Educational Level	**Q275R**	Demographic	The highest level of education attained by the respondent.

Steps for Data Cleaning:

1. Removing the negative values

In the World Values Survey (WVS) dataset, all invalid or missing responses are encoded with negative values instead of being left blank. These values do not represent actual data but are codes for information such as:

- -1, -2, -3, and -5 which represent “Don’t know”, “No answer”, or “Not applicable”.

Those values need to be NA to be later removed because those values do not add any information for the aim of the research and we cause misleading.

2. Recoding Variables

The variables need to be recorded to do analysis on it as they will be defined as character but they really are factors or ordinal factors.

And some of them need to combine their categories and become binary variables because some categories have small (nij) such as:

Original Variable	New Variable	Recoding Action
Q46	**Happiness**	Converted the original 4 categories into a binary factor: "HAPPY" and "NOT happy".
Q6 (Importance of Religion)	**Religion**	Converted the 4-point scale into a binary factor: "Important" and "Not Important".
Q12 (Tolerance)	**Tolerance**	Converted into the factor labels "Yes" and "No".

Original Variable	New Variable	Recoding Action
Q45 (Respect for Authority)	**Authority respect**	Converted the 3-point scale into an Ordinal Factor with labels: "Very interested," "Somewhat," "Not interested", preserving the rank order.
Q50 (Financial Satisfaction)	**Economic satisfaction**	Converted the 10-point scale into a binary factor: "Dissatisfied" and "Satisfied".
Q260 (Sex/Gender)	**Gender**	Converted into "M" and "F".
X003R2 (Age)	**Age**	Converted into ordered categories like "16-29 years," "30-49 years," and "50 and more years."
H_URBRURAL (Settlement Type)	**Region**	Converted into "Urban" and "Rural."
Q275R (Education)	**Education**	Converted into ordered levels like "Lower," "Middle," and "Higher."

3. Handling Missing Values for Final Analysis

This step removes any respondent who has an NA for any of the nine variables used in the final model. This ensures that the regression equation only uses a complete set of data points, a requirement for standard regression techniques.

Descriptive Analysis and Univariate Distributions:

This section provides the descriptive statistics for the final analytical sample (N=2516) after data cleaning and recoding. The objective is to describe the frequency distribution of each categorical variable independently.

1.Univariate Frequency Distributions:

The following analysis details the distribution of the dependent variable and all independent variables, incorporating the final counts and percentages derived from the univariate tables.

A. Dependent Variable: Happiness:

The distribution of the response variable, happiness, is highly skewed.

- An overwhelming majority of respondents (2201, representing 87.5%) categorized themselves as **HAPPY**.
- Only 12.5% (315 respondents) fell into the **NOT happy** category.

This indicates that high happiness levels are the prevailing outcome in the sample.

B. Key Independent Variables:

Variable	Category	Count (n)	Percentage (%)	Interpretation
Region	Urban	2234	88.8%	The sample is heavily concentrated in Urban areas (High Skewness).
	Rural	282	11.2%	This category suffers from critically low representation for robust analysis.
Economical Satisfaction	Satisfied	1551	61.6%	Over three-fifths of the sample express satisfaction, showing a general trend of

Variable	Category	Count (n)	Percentage (%)	Interpretation
				contentment in their household financial status.
	Dissatisfied	965	38.4%	This category is well represented, making it valid for comparison.
Religion	Important	1432	57.0%	Most of the samples consider that religion is important in their life.
	Not Important	1084	43.0%	This category is also strongly represented, which can be used for comparative analysis.
Gender	M (Male)	1356	54.0%	Males constitute a slightly higher share of the sample.
	F (Female)	1160	46.0%	The female representation is high and sufficiently balanced for reliable comparison.
Tolerance	Yes	1760	70.0%	The overwhelming percentage here shows that tolerance is a highly valued social norm and a desirable quality for child development in this sample.
	No	756	30.0%	This minority group is substantial enough to ensure meaningful statistical comparison.
Authority Respect	Very interested	1396	55.5%	**High Commitment:** A majority actively supports the idea of increasing respect for authority or political figures. This represents the modal political preference.

Variable	Category	Count (n)	Percentage (%)	Interpretation
	Somewhat	811	32.2%	This category serves as a buffer between the two extremes.
	Not interested	309	12.3%	This category represents the minority group for this variable.
Age	30-49 years	1051	41.8%	This middle-age group is the most represented cohort.
	50 and more years	856	34.0%	This high percentage ensures strong statistical power when examining age patterns and testing for intergenerational differences in values.
	16-29 years	609	24.2%	This younger group is the least represented.
Education	Higher	1296	51.5%	Over half of the samples reported a higher education level.
	Middle	1176	46.7%	This category is nearly equal to the highest level, resulting in a slightly top-heavy distribution.
	Lower	44	1.8%	This category suffers from **extreme low representation**, posing the most severe constraint on statistical confidence for any findings related to low educational attainment.

2.Key Observations and Data Skewness

The descriptive analysis reveals critical distribution issues that warrant attention in the inferential stages:

High Skewness in Primary Variables: High Skewness in Primary Variables: The variables happiness and region exhibit extreme skewness, with their respective primary categories ("HAPPY" and "Urban") accounting for large percentage of the sample. This limits the power of analysis in the less represented categories.

Low Representation in Categorical Levels:

- **Rural Region:** The extremely low number of rural respondents presents a challenge when using region as a control variable.
- **Lower Education:** The Lower education category has a critically low count of respondents. This minimal representation suggests that any statistical conclusion drawn for this specific group must be treated with significant caution.

Variable Balance: Variables such as Economical Satisfaction and Religion show a balance and are suitable for any coming analysis.

Bivariate Analysis Using Joint and Conditional Probabilities:

In this section, the relationship between happiness and the selected explanatory variables is examined using two–way contingency tables. Joint and conditional probabilities are calculated to describe how the distribution of happiness varies across different demographic and attitudinal groups. This analysis provides an initial descriptive understanding of possible associations before moving to statistical tests in later sections.

1. joint probabilities:

<u>1.1 Happiness and religion importance:</u>

The table shows that respondents who consider religion important report higher happiness levels. Among this group, **52%** are happy compared to **36%** among those who do not consider religion important. This suggests a positive association between valuing religion and happiness.

	happiness		
	HAPPY	NOT happy	Total
religion			
Important	1,300 (52%)	132 (5.2%)	1,432 (57%)
Not Important	901 (36%)	183 (7.3%)	1,084 (43%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

<u>1.2 Happiness and tolerance:</u>

The results show that happiness is higher among individuals who report that children should be taught tolerance at home. In this group, **62% are happy**, compared to **26% among those who say children are not taught tolerance**. This suggests a positive association between promoting tolerance at home and higher levels of happiness.

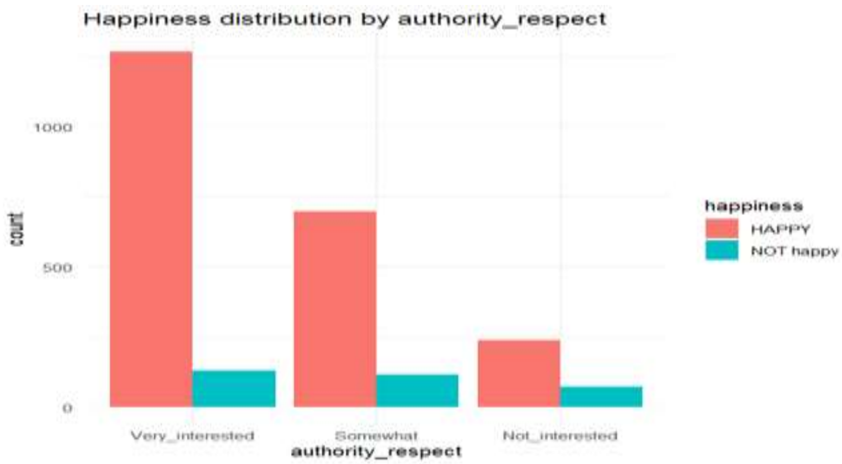
	happiness		
	HAPPY	NOT happy	Total
tolerance			
Yes	1,552 (62%)	208 (8.3%)	1,760 (70%)
No	649 (26%)	107 (4.3%)	756 (30%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

1.3 Happiness and authority respect:

Happiness levels appear to differ depending on how interested respondents are in greater respect for authority. Among those who are *very interested*, **50% report being happy**, compared to **28% among those somewhat interested**, and only **9.5% among those not interested**. This pattern suggests that higher interest in authority

	happiness		
	HAPPY	NOT happy	Total
authority_respect			
Very_interested	1,267 (50%)	129 (5.1%)	1,396 (55%)
Somewhat	696 (28%)	115 (4.6%)	811 (32%)
Not_interested	238 (9.5%)	71 (2.8%)	309 (12%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

as it is also shown from the graph that respect is associated with higher levels of happiness.



1.4 Happiness and economic satisfaction:

It indicates a strong relationship between economic satisfaction and happiness. Among respondents who are economically satisfied, **58% report being happy**, compared to only **29% among those who are dissatisfied**. Additionally, dissatisfaction is associated with a higher percentage of unhappiness (8.9% versus 3.7%).

	happiness		
	HAPPY	NOT happy	Total
economical_satisfication			
Dissatisfied	742 (29%)	223 (8.9%)	965 (38%)
Satisfied	1,459 (58%)	92 (3.7%)	1,551 (62%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

Overall, individuals who feel economically satisfied show noticeably higher levels of happiness.

1.5 Happiness and gender:

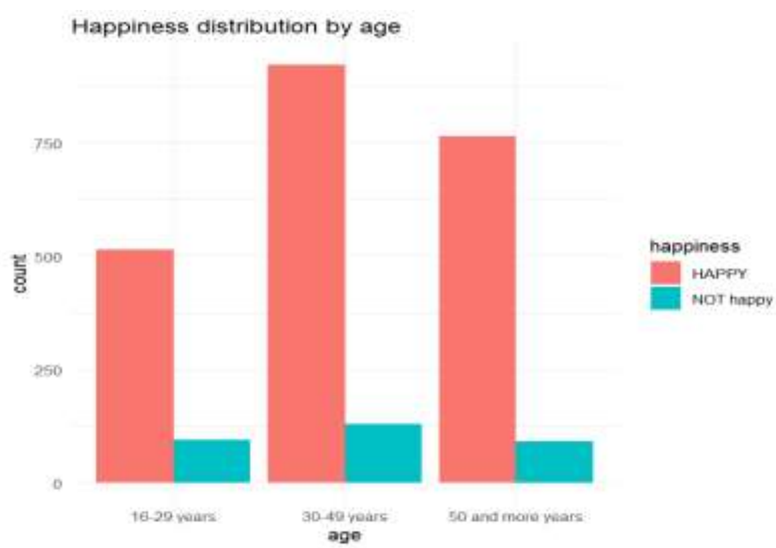
The descriptive results show small differences in the distribution of happiness between men and women. Among men, 47% report being happy, compared to 41% of women. The percentages of those not happy are also close (7.0% vs. 5.6%). These patterns provide **descriptive evidence of slight variation** in happiness levels across gender groups, but no conclusions about association can be made without formal statistical testing.

	happiness		
	HAPPY	NOT happy	Total
gender			
M	1,181 (47%)	175 (7.0%)	1,356 (54%)
F	1,020 (41%)	140 (5.6%)	1,160 (46%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

1.6 Happiness and age categories:

The highest concentration of happy individuals appears in the 30–49 age category (37%), followed by those aged 50 and above (30%). The youngest group, aged 16–29, accounts for 20% of the happy respondents. Although descriptive, this pattern suggests that middle-aged adults represent a substantial portion of the happy population, while younger respondents form the smallest share.

	happiness		
	HAPPY	NOT happy	Total
age			
16-29 years	514 (20%)	95 (3.8%)	609 (24%)
30-49 years	922 (37%)	129 (5.1%)	1,051 (42%)
50 and more years	765 (30%)	91 (3.6%)	856 (34%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)



1.7 Happiness and region:

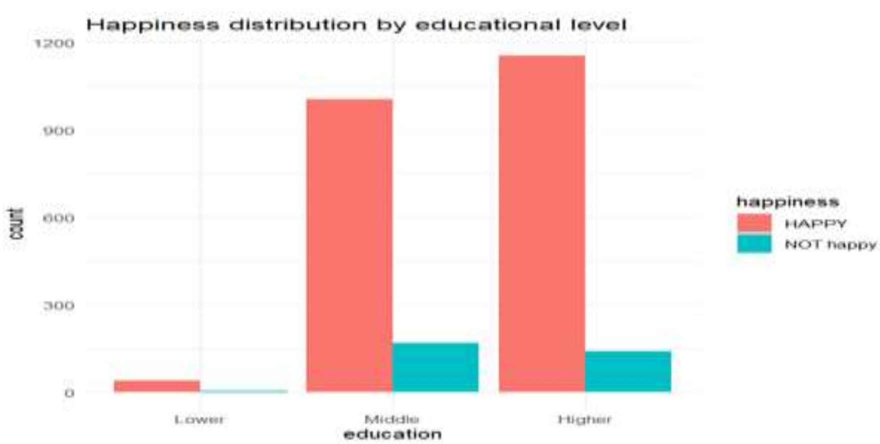
It shows a clear difference in how happiness is spread across urban and rural areas. Most happy respondents reside in urban regions, representing **78%** of all happy individuals, while rural residents account for only **9.8%**. A similar pattern appears among those who are not happy, where **11%** are urban and **1.4%** are rural. These descriptive patterns indicate that happiness is more heavily concentrated among urban respondents, suggesting that urban living conditions may be associated with higher levels of reported happiness within this sample.

	happiness		
	HAPPY	NOT happy	Total
region			
Urban	1,954 (78%)	280 (11%)	2,234 (89%)
Rural	247 (9.8%)	35 (1.4%)	282 (11%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

1.8 Happiness and educational level:

It indicates noticeable variation in happiness across educational groups. The largest share of happy respondents comes from those with higher education, who account for **46%** of all happy individuals. They are followed closely by those with a middle level of education (**40%**), while respondents with lower education represent only **1.6%** of the happy group. A similar pattern appears for the not-happy category, where higher and middle education groups also form the majority.

	happiness		
	HAPPY	NOT happy	Total
education			
Lower	39 (1.6%)	5 (0.2%)	44 (1.7%)
Middle	1,006 (40%)	170 (6.8%)	1,176 (47%)
Higher	1,156 (46%)	140 (5.6%)	1,296 (52%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)



2.Conditional Probabilities (Row Percentages):

2.1 Happiness and religion importance:

When examining the conditional row percentages, happiness appears more frequent among individuals who consider religion important. Within this group, **91% report being happy**, compared with **83%** among those who view religion as not important. Although both groups show high levels of happiness overall, the difference in percentages provides a descriptive indication that valuing religion may be associated with higher reported happiness levels in this sample.

	happiness		
	HAPPY	NOT happy	Total
religion			
Important	1,300 (91%)	132 (9.2%)	1,432 (100%)
Not Important	901 (83%)	183 (17%)	1,084 (100%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

2.2 Happiness and tolerance:

The row percentages show that happiness levels are slightly higher among respondents who say children should be taught tolerance at home. In this group, **88% report being happy**, compared with **86%** in the group that answered "No." Although the difference is small, it provides a descriptive indication that households emphasizing tolerance may show a marginally higher presence of happiness.

	happiness		
	HAPPY	NOT happy	Total
tolerance			
Yes	1,552 (88%)	208 (12%)	1,760 (100%)
No	649 (86%)	107 (14%)	756 (100%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

2.3 Happiness and authority respect:

It shows a clear gradient in happiness across levels of interest in greater respect for authority. Happiness is highest among those who are *very interested*, with **91% reporting happiness**. This percentage decreases to **86%** among those who are *somewhat interested*, and further to **77%** among respondents who are *not interested*. These descriptive patterns indicate that higher interest in authority and respect corresponds with a larger share of individuals reporting happiness in this sample.

	happiness		
	HAPPY	NOT happy	Total
authority_respect			
Very_interested	1,267 (91%)	129 (9.2%)	1,396 (100%)
Somewhat	696 (86%)	115 (14%)	811 (100%)
Not_interested	238 (77%)	71 (23%)	309 (100%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

2.4 Happiness and economic satisfaction:

The row percentages show a substantial difference in happiness between individuals who are satisfied and those who are dissatisfied with their economic situation. Among the satisfied group, **94% report being happy**, compared with **77%** among those who are dissatisfied. Meanwhile, unhappiness is more concentrated among the dissatisfied group (23% versus 5.9%). These descriptive results indicate that higher economic satisfaction coincides with a noticeably greater share of individuals reporting happiness.

	happiness		
	HAPPY	NOT happy	Total
economical_satisfication			
Dissatisfied	742 (77%)	223 (23%)	965 (100%)
Satisfied	1,459 (94%)	92 (5.9%)	1,551 (100%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

2.5 Happiness and gender:

It shows very small differences in happiness between men and women. Among men, **87% report being happy**, compared with **88%** among women. The proportions of not-happy respondents are also nearly identical (13% vs. 12%). Overall, the distribution suggests minimal variation in happiness across gender groups within this sample.

	happiness		
	HAPPY	NOT happy	Total
gender			
M	1,181 (87%)	175 (13%)	1,356 (100%)
F	1,020 (88%)	140 (12%)	1,160 (100%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

2.6 Happiness and age categories:

The row percentages show a slight upward trend in happiness with increasing age. Among respondents aged 16–29, **84% report being happy**, rising to **88%** in the 30–49 group, and reaching **89%** among those aged 50 and above. The share of not-happy individuals decreases gradually across these age categories. These descriptive patterns suggest that older respondents tend to report happiness at marginally higher rates than younger ones.

	happiness		
	HAPPY	NOT happy	Total
age			
16-29 years	514 (84%)	95 (16%)	609 (100%)
30-49 years	922 (88%)	129 (12%)	1,051 (100%)
50 and more years	765 (89%)	91 (11%)	856 (100%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

2.7 Happiness and region:

show almost identical happiness levels across urban and rural areas. Among urban respondents, **87% report being happy**, while **88%** of rural respondents report happiness. The share of not-happy individuals is similarly close (13% vs. 12%). These descriptive patterns indicate minimal variation in reported happiness between urban and rural regions in this sample.

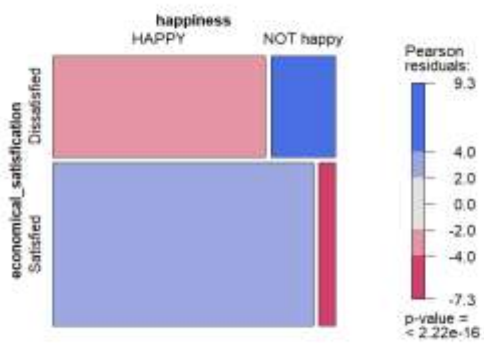
	happiness		
	HAPPY	NOT happy	Total
region			
Urban	1,954 (87%)	280 (13%)	2,234 (100%)
Rural	247 (88%)	35 (12%)	282 (100%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

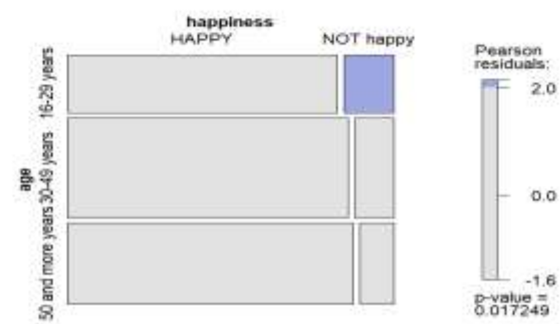
2.8 Happiness and educational level:

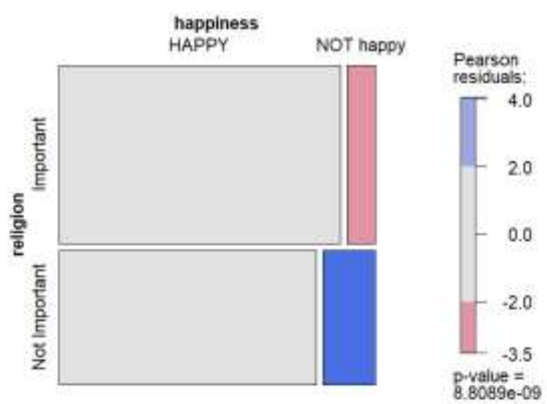
Happiness is consistently high across all educational levels. Among those with lower education, 89% report being happy and 11% not happy; for middle education, 86% are happy and 14% not happy; and for higher education, 89% are happy and 11% not happy. These row percentages indicate that educational level does not appear to have a strong influence on happiness in this dataset.

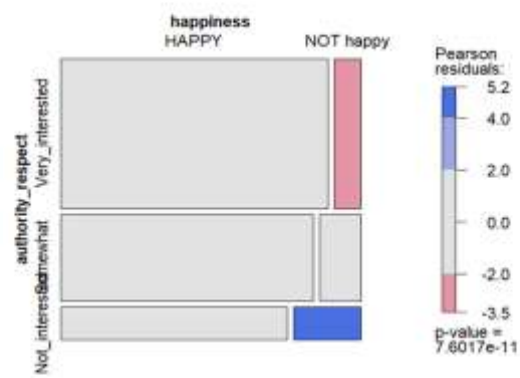
	happiness		
	HAPPY	NOT happy	Total
education			
Lower	39 (89%)	5 (11%)	44 (100%)
Middle	1,006 (86%)	170 (14%)	1,176 (100%)
Higher	1,156 (89%)	140 (11%)	1,296 (100%)
Total	2,201 (87%)	315 (13%)	2,516 (100%)

3. Graphical representation and more indications:

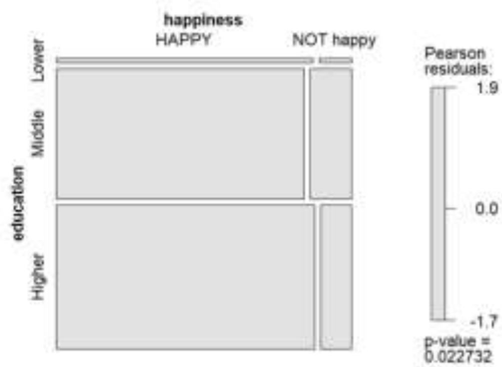


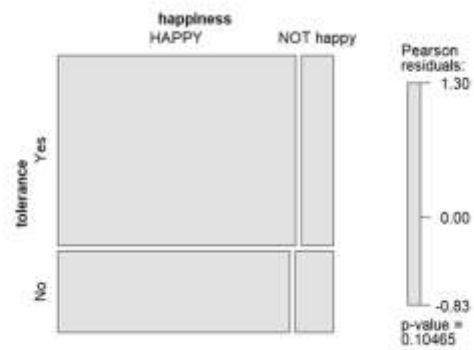


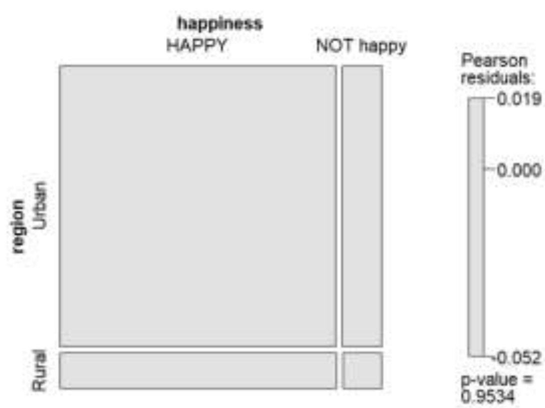


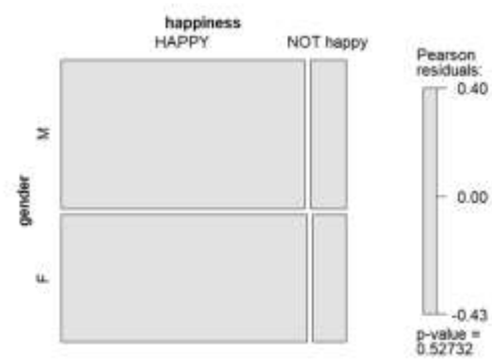


These mosaic graphs for the distribution of happiness through (“economic satisfaction “,” authority respect”,” religion”,” age”) show a great indication for dependency association that will be detected by different tests in the following sections.









On the other hand, we can have an indication that the mosaic plot for (“education”,” tolerance”,” region “, “gender”) and happiness doesn’t have an association it also can’t be surely assumed without testing.

Analysis Using Odds Ratios, Chi-Square Tests, and Measures of Association:

To present an analytical summary of the categorical variables included in the data and their relationship with overall happiness. The analysis will rely on odds ratios for binary variables, Chi-Square tests of independence, and measures of association.

1. Odds Ratios for Binary Variables:

1.1 Region (Urban vs. Rural)

```
$measure
       odds ratio with 95% C.I.
        estimate     lower    upper
  Urban 1.0000000        NA       NA
  Rural 0.9923879 0.6711928 1.427088

$p.value
       two-sided
        midp.exact fisher.exact chi.square
  Urban         NA           NA         NA
  Rural  0.9682029            1  0.9533977
```

The odds ratio analysis for the relationship between **region** and **happiness** shows that the estimated odds ratio for rural respondents compared to urban respondents is **0.99** (95% CI: **0.67–1.43**). Since the confidence interval includes 1, this indicates **no meaningful difference** in the likelihood of being happy between individuals living in urban and rural areas. Additionally, the p-value (**p ≈ 0.97**) confirms that this association is **statistically insignificant**.
Therefore, as shown in the output, **region does not appear to influence happiness levels** in this dataset.

1.2 Religion Importance

```
$measure
              odds ratio with 95% C.I.
               estimate  lower    upper
  Important     1.000000     NA       NA
  Not Important 1.999096 1.5748 2.543881

$p.value
              two-sided
                 midp.exact fisher.exact   chi.square
  Important              NA           NA           NA
  Not Important 1.114168e-08 1.253898e-08 8.808913e-09
```

The odds ratio analysis examining the association between **religion** and **happiness** indicates a statistically significant relationship. Individuals who consider religion **"Not Important"** have an odds ratio of **2.00** (95% CI: **1.57–2.54**) relative to those who consider it important.

This means they are **twice as likely to report being NOT happy** compared to those who view religion as important.
Because the confidence interval does **not** include 1 and the p-values from all tests are extremely low (**p < 0.00000002**), the association is **highly statistically significant**.

Overall, as shown in the output, **religion importance is a strong predictor of happiness**, with lower importance associated with a higher likelihood of unhappiness.

1.3 Tolerance

```
$measure
     odds ratio with 95% C.I.
      estimate     lower     upper
  Yes 1.000000        NA        NA
  No  1.230851 0.9552799 1.578109

$p.value
     two-sided
      midp.exact fisher.exact chi.square
  Yes         NA           NA         NA
  No   0.1074875    0.1147107  0.1046505
```

The odds ratio analysis examining the association between **tolerance** and **happiness** indicates a statistically non-significant relationship. Individuals who answered **"No"** have an odds ratio of **1.23** (95% CI: 0.96–1.58) relative to those who answered **"Yes"**.

This means that the "No" group has slightly higher odds of being **NOT happy** compared to the "Yes" group. However, because the confidence interval **includes 1** and the p-values from all tests are greater than 0.05 (mid-p = 0.107, Fisher's exact = 0.115, chi-square = 0.105), the association is **not statistically significant**.

Overall, as shown in the output, tolerance does **not appear to be a strong predictor of happiness** in the data.

1.4 Economic Satisfaction

```
$measure
             odds ratio with 95% C.I.
               estimate     lower     upper
  Dissatisfied 1.0000000        NA        NA
  Satisfied    0.2101522 0.1615842 0.2712629

$p.value
             two-sided
              midp.exact fisher.exact   chi.square
  Dissatisfied        NA           NA           NA
  Satisfied            0 1.107742e-35 9.934373e-37
```

The odds ratio analysis examining the association between **economical satisfaction** and **happiness** indicates a clear and statistically significant relationship. Individuals who are **"Satisfied"** with their economic situation have an odds ratio of **0.21** (95% CI: 0.16–0.27) relative to those who are **“Dissatisfied”**.

This means that there is a significant association between economic satisfaction and happiness: satisfied individuals are less likely to report being NOT happy compared to dissatisfied individuals.

The confidence interval does **not** include 1, and the p-values from all tests are extremely low (mid-p = 0, Fisher’s exact ≈ 1.1e-35, chi-square ≈ 9.9e-37), indicating that the association is **statistically significant**.

Overall, as shown in the output, **economic satisfaction is associated with happiness**, with dissatisfaction associated with a higher likelihood of unhappiness.

<u>1.5 Gender</u>

The odds ratio analysis examining the association between **gender** and **happiness** suggests **no statistically significant relationship**. Females (F) have an odds ratio of **0.93** (95% CI: 0.73–1.17) compared to males (M), who are used as the reference group.

```
$measure
   odds ratio with 95% C.I.
     estimate     lower    upper
  M 1.0000000        NA       NA
  F 0.9264906 0.7297392 1.174358

$p.value
   two-sided
    midp.exact fisher.exact chi.square
  M         NA           NA         NA
  F  0.5285856    0.5459877  0.5273203
```

This indicates that there is **no clear association** between gender and the likelihood of being happy or not. The confidence interval **includes 1**, and all p-values from the tests (mid-p = 0.529, Fisher’s exact = 0.546, chi-square = 0.527) are greater than 0.05, confirming that the relationship is **not statistically significant**.

Overall, as shown in the output, **happiness appears to be independent of gender** in this dataset.

2. Chi-Square Tests of Independence:

This section presents Chi-Square tests of independence conducted for variables that demonstrated a potential association with happiness in the odds ratio analysis. Variables found to be independent in the odds ratio assessment are **excluded**, as further testing would not yield additional insights. These tests provide formal statistical confirmation of the associations suggested by the earlier analysis.

H_0 (Null Hypothesis):

The two variables are independent

H_1 (Alternative Hypothesis):

The two variables are not independent

2.1 Religion Importance

The Chi-Square test of independence was performed to formally examine the relationship between **religion** and **happiness**. The test yielded **$X^2 = 33.088$** with **1 degree of freedom** and a **p-value of 8.81×10^{-9}**.

```
· chisq.test(table2,correct=FALSE)

        Pearson's Chi-squared test

lata:  table2
(-squared = 33.088, df = 1, p-value = 8.809e-09
```

Given that the p-value is far below the conventional 0.05 threshold, we **reject the null hypothesis** of independence. This provides strong statistical evidence that **religion importance and happiness are associated**, supporting the findings from the earlier odds ratio analysis.

Overall, as illustrated in Table 2 and confirmed by this test, individuals who consider religion less important are more likely to report being NOT happy.

2.2 Economic Satisfaction

The Chi-Square test of independence was conducted to assess the relationship between **economic satisfaction** and **happiness**. The test produced **X² = 160.26** with **1 degree of freedom** and a **p-value < 2.2 × 10⁻¹⁶**.

```
> chisq.test(table6,correct=FALSE)

	Pearson's Chi-squared test

data:  table6
X-squared = 7.5679, df = 2, p-value = 0.02273
```

Since the p-value is extremely small, we **reject the null hypothesis** of independence. This indicates a **highly significant association** between economic satisfaction and happiness, confirming the association observed in the odds ratio analysis.

Overall, the results show that individuals who are dissatisfied with their economic situation are significantly more likely to report being NOT happy, as reflected in the contingency table and statistical tests.

2.3 Education Level

The Chi-Square test of independence was conducted to examine the relationship between **education level** and **happiness**. The test yielded **X² = 7.568** with **2 degrees of freedom** and a **p-value = 0.0227**.

```
> chisq.test(table4,correct=FALSE)

	Pearson's Chi-squared test

data:  table4
X-squared = 160.26, df = 1, p-value < 2.2e-16
```

Since the p-value is below the conventional 0.05 threshold, we **reject the null hypothesis** of independence. This indicates a **statistically significant association** between education level and happiness.

Overall, as observed in the contingency table, **education appears to influence happiness**, with variation in happiness levels across different educational categories.

2.4 Age

The Chi-Square test of independence was performed to evaluate the association between **age** and **happiness**. The test produced **X^2 = 8.12** with **2 degrees of freedom** and a **p-value = 0.0173**.

```
> chisq.test(table8,correct=FALSE)

        Pearson's Chi-squared test

data:  table8
X-squared = 46.6, df = 2, p-value = 7.602e-11
```

Since the p-value is below the 0.05 significance level, we **reject the null hypothesis** of independence. This provides evidence of a **statistically significant association** between age and happiness.

Overall, as shown in the contingency table, **happiness levels vary across age groups**, suggesting that age may play a role in influencing reported happiness.

2.5 Authority Respect

The Chi-Square test of independence was conducted to assess the relationship between **respect** and **happiness**. The test yielded **X^2 = 46.6** with **2 degrees of freedom** and a **p-value = 7.60×10^{-11}**.

```
> chisq.test(table7,correct=FALSE)

        Pearson's Chi-squared test

data:  table7
X-squared = 8.12, df = 2, p-value = 0.01725
```

Given the extremely low p-value, we **reject the null hypothesis** of independence. This provides strong statistical evidence of a **significant association** between respect for authority and happiness.

Overall, as illustrated in the contingency table, **levels of happiness differ considerably across categories of authority respect**, highlighting the role of this variable in influencing happiness outcomes.

3. Measures of Association

3.1 Religion

To further quantify the relationship between **religion's importance** and **happiness**, several measures of association were computed.

- **Cramer's V = 0.115** and **Contingency Coefficient = 0.114** indicate a **weak but positive association** between the two variables.
- **Kendall Tau-b = 0.115** and **Spearman Correlation = 0.115** confirm a **low positive monotonic relationship**.
- **Goodman-Kruskal Gamma = 0.333** suggests a moderate directional association, indicating that individuals who consider religion "Not Important" are somewhat more likely to report being unhappy.
- Other measures (Somers' D, Stuart Tau-c) consistently support a weak positive association.

Overall, as seen from the output, while there is a **statistically significant relationship** between religion importance and happiness (from the Chi-Square test), the **strength of the association is relatively weak**, suggesting that other factors may also influence happiness.

```
> DescTools::Assocs(table2, conf.level = 0.95, verbose = NULL)
                          estimate  lwr.ci  upr.ci
Contingency Coeff.          0.1139       -       -
Cramer V                    0.1147  0.0756  0.1538
Kendall Tau-b               0.1147  0.0747  0.1546
Goodman Kruskal Gamma       0.3334  0.2270  0.4398
Stuart Tau-c                0.0752  0.0488  0.1015
Somers D C|R                0.0766  0.0498  0.1035
Somers D R|C                0.1716  0.1064  0.2368
Pearson Correlation         0.1147  0.0759  0.1531
Spearman Correlation        0.1147  0.0759  0.1531
Lambda C|R                  0.0000  0.0000  0.0000
Lambda R|C                  0.0470  0.0157  0.0784
Lambda sym                  0.0365  0.0122  0.0607
Uncertainty Coeff. C|R      0.0173  0.0055  0.0290
Uncertainty Coeff. R|C      0.0095  0.0030  0.0160
Uncertainty Coeff. sym      0.0123  0.0039  0.0206
Mutual Information          0.0094       -       -
```

3.2 Economic Satisfaction

The association measures for **economic satisfaction** and **happiness** indicate a **moderate negative relationship**:

- **Cramer's V = 0.252** and **Contingency Coefficient = 0.245** suggest a moderate association between economic satisfaction and happiness.
- **Kendall Tau-b = -0.252** and **Pearson/Spearman correlations = -0.252** indicate that higher economic satisfaction is associated with higher likelihood of being happy (negative sign reflects coding direction).
- **Goodman-Kruskal Gamma = -0.653** shows a relatively strong directional association, meaning that dissatisfied individuals are much more likely to report being unhappy.
- Other measures, including Somers' D and Stuart Tau-c, support this negative association consistently.

Overall, while the odds ratio and chi-square test already indicate a statistically significant association, these measures quantify the **strength and direction**, showing that **economic dissatisfaction is substantially linked with lower happiness**.

```
> DescTools::Assocs(table4, conf.level = 0.95, verbose = NULL)
                          estimate  lwr.ci  upr.ci
Contingency Coeff.          0.2447       -       -
Cramer V                    0.2524  0.2133  0.2915
Kendall Tau-b              -0.2524 -0.2940 -0.2108
Goodman Kruskal Gamma      -0.6531 -0.7272 -0.5791
Stuart Tau-c               -0.1625 -0.1901 -0.1348
Somers D C|R               -0.1718 -0.2008 -0.1427
Somers D R|C               -0.3708 -0.4675 -0.2741
Pearson Correlation        -0.2524 -0.2886 -0.2154
Spearman Correlation       -0.2524 -0.2886 -0.2154
Lambda C|R                  0.0000  0.0000  0.0000
Lambda R|C                  0.1358  0.1022  0.1693
Lambda sym                  0.1023  0.0776  0.1271
Uncertainty Coeff. C|R      0.0824  0.0578  0.1069
Uncertainty Coeff. R|C      0.0467  0.0322  0.0611
Uncertainty Coeff. sym      0.0596  0.0415  0.0777
Mutual Information          0.0448       -       -
```

3.3 Education Level

The measures of association between education level and happiness indicate a **very weak negative relationship**.

- **Cramer's V = 0.055** and **Contingency Coefficient = 0.055** suggest that the association is minimal.
- **Kendall's Tau-b = -0.051)** and **Pearson/Spearman correlations around -0.05** confirm a very weak negative trend.
- **Goodman-Kruskal Gamma = -0.151** shows a slight ordinal association.
- Other measures, including Somers' D, Stuart Tau-c, and the uncertainty coefficients, consistently indicate a weak and nearly negligible association.

Overall, Education level is related to happiness in the data, but the strength of the association is weak. This suggests that other factors likely play a larger role in determining happiness, while education has only a minor influence.

```
> DescTools::Assocs(table6, conf.level = 0.95, verbose = NULL)
                          estimate  lwr.ci  upr.ci
:ontingency Coeff.          0.0548       -       -
:ramer V                    0.0548  0.0046  0.0911
(endall Tau-b              -0.0511 -0.0898 -0.0125
;oodman Kruskal Gamma      -0.1510 -0.2630 -0.0390
;tuart Tau-c               -0.0344 -0.0604 -0.0084
;omers D C|R               -0.0333 -0.0585 -0.0081
;omers D R|C               -0.0785 -0.1451 -0.0118
'earson Correlation        -0.0489 -0.0878 -0.0099
;pearman Correlation       -0.0516 -0.0904 -0.0125
.ambda C|R                  0.0000  0.0000  0.0000
.ambda R|C                  0.0246  0.0000  0.0525
.ambda sym                  0.0195  0.0000  0.0418
Jncertainty Coeff. C|R      0.0040 -0.0017  0.0096
Jncertainty Coeff. R|C      0.0020 -0.0008  0.0047
Jncertainty Coeff. sym      0.0026 -0.0011  0.0064
'utual Information          0.0022       -       -
```

3.4 Age

The measures of association between age and happiness indicate a very weak negative relationship.

- **Contingency Coefficient = 0.057** and **Cramer's V = 0.057**.
- **Kendall's Tau-b = -0.052** (95% CI: -0.089 to -0.014) indicates a very slight negative ordinal association, showing that as age increases, there is a minimal tendency toward lower happiness.
- **Pearson Correlation = -0.055** and **Spearman Correlation = -0.055** confirm that both linear and monotonic relationships are extremely weak.
- **Goodman-Kruskal Gamma = -0.136** reflects a minor negative ordinal association, indicating that age has only a small impact on happiness ranking.
- Other measures, including **Somers' D, Stuart Tau-c = -0.039**, and the **uncertainty coefficients**, consistently indicate a negligible association between age and happiness.

Overall, these results suggest that age has minimal influence on happiness in the data.

```
> DescTools::Assocs(table7, conf.level = 0.95, verbose = NULL)
                         estimate  lwr.ci  upr.ci
Contingency Coeff.         0.0567       -       -
Cramer V                   0.0568  0.0091  0.0931
Kendall Tau-b             -0.0518 -0.0893 -0.0144
Goodman Kruskal Gamma     -0.1359 -0.2326 -0.0392
Stuart Tau-c              -0.0392 -0.0675 -0.0108
Somers D C|R              -0.0301 -0.0518 -0.0083
Somers D R|C              -0.0894 -0.1560 -0.0228
Pearson Correlation       -0.0554 -0.0943 -0.0164
Spearman Correlation      -0.0548 -0.0937 -0.0158
Lambda C|R                 0.0000  0.0000  0.0000
Lambda R|C                 0.0000  0.0000  0.0000
Lambda sym                 0.0000  0.0000  0.0000
Uncertainty Coeff. C|R     0.0042 -0.0017  0.0100
Uncertainty Coeff. R|C     0.0015 -0.0006  0.0035
Uncertainty Coeff. sym     0.0022 -0.0009  0.0052
Mutual Information         0.0023       -       -
```

3.5 Respect for Authority

The measures of association between respect for authority and happiness indicate a weak positive relationship.

- **Contingency Coefficient = 0.135** and **Cramer's V = 0.136** suggest a modest association.
- **Kendall's Tau-b = 0.122** (95% CI: 0.082–0.163) indicates a slight positive ordinal association, showing that higher respect for authority tends to correspond to higher happiness.
- **Pearson Correlation = 0.134** and **Spearman Correlation = 0.127** confirm a weak linear and monotonic relationship.
- **Goodman-Kruskal Gamma = 0.317** reflects a moderate positive ordinal association, further supporting the trend that greater respect for authority is associated with greater happiness.
- Other measures, including **Somers' D Stuart Tau-c = 0.087**, and the **uncertainty coefficients**, consistently indicate a positive, albeit weak, association between authority respect and happiness.

Overall, these results suggest that respect for authority is positively associated with happiness, though the strength of the association is modest.

```
> DescTools::Assocs(table8, conf.level = 0.95, verbose = NULL)
                          estimate  lwr.ci  upr.ci
Contingency Coeff.          0.1349       -       -
Cramer V                    0.1361  0.0953  0.1739
Kendall Tau-b               0.1224  0.0822  0.1625
Goodman Kruskal Gamma       0.3170  0.2249  0.4090
Stuart Tau-c                0.0867  0.0580  0.1154
Somers D C|R                0.0756  0.0507  0.1006
Somers D R|C                0.1979  0.1217  0.2741
Pearson Correlation         0.1338  0.0953  0.1720
Spearman Correlation        0.1273  0.0887  0.1656
Lambda C|R                  0.0000  0.0000  0.0000
Lambda R|C                  0.0000  0.0000  0.0000
Lambda sym                  0.0000  0.0000  0.0000
Uncertainty Coeff. C|R      0.0224  0.0086  0.0362
Uncertainty Coeff. R|C      0.0089  0.0034  0.0144
Uncertainty Coeff. sym      0.0127  0.0048  0.0206
Mutual Information          0.0122       -       -
```

Then from all the previous results we can conclude that Economic satisfaction is the strongest predictor of happiness, followed by religion importance and authority respect. Education and age show weak associations, while region, gender, and tolerance show none.

Three-way Contingency tables:

In this part of the analysis, we construct **three-way contingency tables** to explore the relationships between pairs of categorical variables while **controlling for a third variable**, such as region or gender. For example, we examine the association between **happiness and age** within each level of **region** (Urban vs. Rural), or between **happiness and education** while fixing **gender** (Male vs. Female). This approach allows us to investigate how the relationship between two variables **changes across different levels of a third variable**, providing deeper insight into potential interactions and patterns in the data. By summarizing the counts and percentages within each layer of the table, we can identify trends, compare distributions, and understand multidimensional categorical relationships in a clear and structured manner. In addition to constructing three-way contingency tables, we will create **mosaic plots** to visually represent the relationships between categorical variables. This graphical approach complements the tables by providing an intuitive visualization of patterns, distributions, and potential associations in the data.

1.Three-way contingency tables

1.1 Happiness, Age and Region:

Overall, most respondents are happy in both Urban (≈87.4%) and Rural (≈87.6%) areas. In Urban regions, happiness peaks in the **30–49 age group**, while in Rural areas, it peaks among **older adults (50+)**.

$Urban

happiness/age	16-29 years	30-49 years	50 and more years	Total
HAPPY	471	828	655	1954
NOT happy	86	110	84	280
Total	557	938	739	2234

$Rural

happiness/age	16-29 years	30-49 years	50 and more years	Total
HAPPY	43	94	110	247
NOT happy	9	19	7	35
Total	52	113	117	282

The proportion of “NOT happy” respondents is low across all age groups (<18%), but slightly higher among younger and middle-aged Rural adults. These results indicate that **age and region influence happiness**, with Urban happiness highest in middle age and Rural happiness highest in older age.

1.2 Happiness, Economical_satisfication and Region

In both Urban and Rural areas, economic satisfaction is strongly associated with happiness. Among HAPPY respondents, the majority are economically satisfied (≈66–65%), whereas most NOT happy respondents are economically dissatisfied (≈70% Urban, 80% Rural). This indicates that **economic satisfaction plays a significant role in perceived happiness**, with the effect being more pronounced in Rural areas.

$Urban

happiness/economical_satisfication	Dissatisfied	Satisfied	Total
HAPPY	655	1299	1954
NOT happy	195	85	280
Total	850	1384	2234

$Rural

happiness/economical_satisfication	Dissatisfied	Satisfied	Total
HAPPY	87	160	247
NOT happy	28	7	35
Total	115	167	282

1.3 Happiness, Education and Region:

In both Urban and Rural areas, most happy respondents have **middle or higher education**, whereas lower-educated individuals form a very small portion of happy respondents (≈1–5%). NOT happy respondents are also concentrated in middle and higher education groups, with lower education almost negligible. This suggests that **education level is positively associated with happiness**, with higher education generally linked to greater reported happiness.

$Urban

happiness/education	Lower	Middle	Higher	Total
HAPPY	28	878	1048	1954
NOT happy	5	150	125	280
Total	33	1028	1173	2234

$Rural

happiness/education	Lower	Middle	Higher	Total
HAPPY	11	128	108	247
NOT happy	0	20	15	35
Total	11	148	123	282

1.4 Happiness, Authority respect and Gender:

For both males and females, the level of interest in authority and respect is positively associated with happiness. Individuals who are **very interested** in authority/respect report the highest happiness rates (≈90–91%), whereas those less interested show lower happiness percentages (≈75–80%). This pattern is consistent across genders, indicating that **greater respect for authority is linked to higher self-reported happiness**.

```
$M
happiness/authority_respect Very_interested Somewhat Not_interested Total
                      HAPPY             661      373            147  1181
                  NOT happy              70       56             49   175
                      Total             731      429            196  1356

$F
happiness/authority_respect Very_interested Somewhat Not_interested Total
                      HAPPY             606      323             91  1020
                  NOT happy              59       59             22   140
                      Total             665      382            113  1160
```

1.5 Happiness, Religion and Gender:

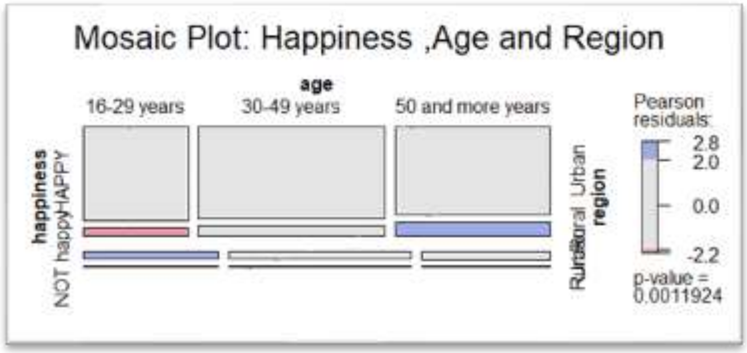
For both males and females, considering religion as important is associated with higher happiness. Among males, 640 out of 705 (≈90.8%) of those who consider religion important are happy, compared to 541 out of 651 (≈83.1%) who consider it not important. Similarly, among females, 660 out of 727 (≈90.8%) of the Important group are happy, compared to 360 out of 433 (≈83.1%) in the Not Important group. This indicates that **religion plays a positive role in perceived happiness** across genders.

```
$M
happiness/religion Important Not Important Total
             HAPPY       640           541  1181
         NOT happy        65           110   175
             Total       705           651  1356

$F
happiness/religion Important Not Important Total
             HAPPY       660           360  1020
         NOT happy        67            73   140
             Total       727           433  1160
```

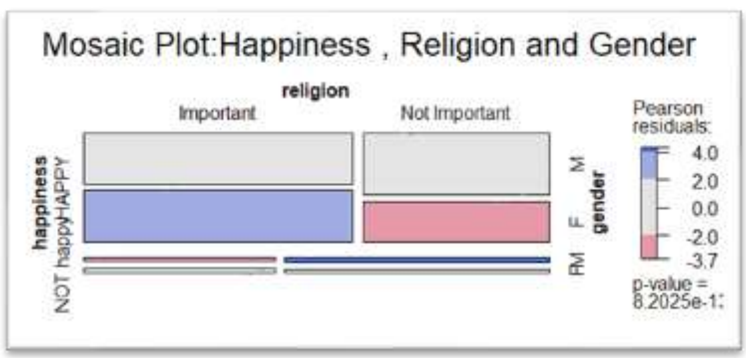
2.Mosaic plots:

2.1Happiness, Age and Region:



The mosaic plot shows clear deviations from what we would expect under independence, indicating that **happiness is associated with both age and region**. Younger individuals tend to be happier than expected, while older adults (especially those living in **rural areas**) show higher levels of unhappiness. Middle-aged groups display relatively neutral patterns, suggesting weaker association. The colored residuals highlight that the variables do not behave independently, which aligns with the very small p-value and supports rejecting the null hypothesis of independence.

2.2Happiness, Religion and Gender:



The mosaic plot shows that **happiness is strongly associated with both religion importance and gender**, rather than being independent. Individuals who consider religion *important* show noticeably **higher happiness than expected**, especially among females, as indicated by the strong blue shading. In contrast, those who view religion as *not important* show **lower-than-expected happiness**, particularly among females, shown by the red shading. The clear residual patterns and extremely small p-value suggest that happiness, religion importance, and gender interact in a meaningful way, and the variables **do not behave independently**.

3.Assessing the Associations Between Categorical Variables:

Testing Conditional Independence Among Categorical Variables.

a. **Using Chi-square test:**

A chi-square test of independence was conducted. Fewer than 20% of the counts were below 5, so the test assumptions were met.

Testing Conditional Independence Among Happiness, Age controlling for region

<u>3.1 Happiness, Age and Region</u>

H_0 (Null Hypothesis):

Age is conditionally independent of Happiness given Region

H_1 (Alternative Hypothesis):

Age is NOT conditionally independent of Happiness given Region.

$$df_{\text{overall}} = df_{\text{Urban}} + df_{\text{Rural}}$$
$$\chi^2_{\text{overall}} = 5.7659 + 7.6094 = 13.3753$$
$$df_{\text{overall}} = 2 + 2 = 4$$

```
> x

        Pearson's Chi-squared test

data:  table_3way11$Urban
X-squared = 5.7659, df = 2, p-value = 0.05597

> y

        Pearson's Chi-squared test

data:  table_3way11$Rural
X-squared = 7.6094, df = 2, p-value = 0.02227
```

Tabulated chi-square value: $\chi^2_{0.05,4} = 9.488$

$$\chi^2_{\text{overall}} = 13.3753 > 9.488$$

- Reject the null hypothesis at 5% significance level, Age is not conditionally independent of Happiness given Region (When considering both Urban and Rural regions together, the overall chi-square test indicates that Age is not conditionally independent of Happiness given Region ($\chi^2 = 13.38$, df = 4, $p < 0.05$).

3.2 Happiness, Education and Region

H₀ (Null Hypothesis):

Education is conditionally independent of Happiness given Region

H₁ (Alternative Hypothesis):

Education is NOT conditionally independent of Happiness given Region

$$\chi^2_{\text{overall}} = 7.9477 + 7.9477 = 15.8954$$

Tabulated chi-square value: $\chi^2_{0.05,4} = 9.488$

$\chi^2_{\text{overall}} = 15.8954 > 9.488$

➢ Reject the null hypothesis at 5% significance level, Education is **not conditionally independent** of Happiness given Region (When considering both Urban and Rural regions together, the overall chi-square test indicates that Education is **not conditionally independent** of Happiness given Region ($\chi^2 = 15.89$, df = 4, $p < 0.05$).

```
> a

        Pearson's Chi-squared test

data:  table_3way22$Urban
X-squared = 7.9477, df = 2, p-value = 0.0188

> b

        Pearson's Chi-squared test

data:  table_3way22$Urban
X-squared = 7.9477, df = 2, p-value = 0.0188
```

3.3 Happiness, Authority respect and Gender

H₀ (Null Hypothesis):

Authority respect is conditionally independent of Happiness given Gender

H₁ (Alternative Hypothesis):

Authority respect is NOT conditionally independent of Happiness given Gender

$$\chi^2_{\text{overall}} = 32.725 + 16.337 = 49.062$$

Tabulated chi-square value: $\chi^2_{0.05,4} = 9.488$

$\chi^2_{\text{overall}} = 49.062 > 9.488$

➢ Reject the null hypothesis at 5% significance level, **Authority respect** is

```
> s

        Pearson's Chi-squared test

data:  table_3way55$M
X-squared = 32.725, df = 2, p-value = 7.83e-08

> m

        Pearson's Chi-squared test

data:  table_3way55$F
X-squared = 16.337, df = 2, p-value = 0.0002834
```

not conditionally independent of Happiness given Gender (When considering both Male and Female groups together, the overall chi-square test indicates that **Authority respect** is not conditionally independent of Happiness given Gender ($\chi^2 = 49.06$, df = 4, $p < 0.05$).

4.Describing the Association Using Relative Risk:

4.1 Happiness, Economical_Satisfication and Region

```
$measure
          risk ratio with 95% C.I.
happiness   estimate    lower    upper
  HAPPY      1.00000       NA       NA
  NOT happy  2.07759 1.881021 2.294701
```

```
$measure
          risk ratio with 95% C.I.
happiness   estimate    lower    upper
  HAPPY     1.000000       NA       NA
  NOT happy 2.271264 1.792491 2.877918
```

Urban Areas:	Rural Areas:
RR for NOT happy vs HAPPY: 2.08 (95% CI: 1.88 – 2.29), NOT happy individuals are about 2 times more likely to report dissatisfaction compared to HAPPY individuals in urban areas.	RR for NOT happy vs HAPPY: 2.27 (95% CI: 1.79 – 2.88), NOT happy individuals are more than twice as likely to report dissatisfaction compared to HAPPY individuals in rural areas.

- The RR is slightly higher in rural areas (2.27) than urban areas (2.08), suggesting the impact of unhappiness on dissatisfaction may be stronger in rural contexts.

4.2 Happiness, Religion and Gender

```
$measure
            risk ratio with 95% C.I.
happiness     estimate     lower     upper
  HAPPY      1.0000000        NA        NA
  NOT happy 0.6854018 0.5613031 0.8369375
```

Males:

```
$measure
            risk ratio with 95% C.I.
happiness     estimate     lower     upper
  HAPPY      1.0000000        NA        NA
  NOT happy 0.7396104 0.6185491 0.8843655
```

Females:

R for NOT happy vs HAPPY: 0.69 (95% CI: 0.56 – 0.84), Among males, NOT happy individuals are about 31% less likely to consider religion “Important” compared to HAPPY individuals.	RR for NOT happy vs HAPPY: 0.74 (95% CI: 0.62 – 0.88), Among females, NOT happy individuals are 26% less likely to consider religion “Important” compared to HAPPY individuals.

- The negative association between unhappiness and considering religion important is slightly weaker among females (RR = 0.74) than males (RR = 0.69**).**

b. Using Cochran–Mantel–Haenszel (CMH) test:

The Cochran–Mantel–Haenszel (CMH) test examines the association between two categorical 2×2 variables while controlling for a third stratifying variable. It tests whether the variables are conditionally independent across strata, assuming that the odds ratios are homogeneous (similar) in all strata.

5.1 Happiness, Economical_Satisfication and Region

H_0 (Null Hypothesis):

Economic Satisfaction is conditionally independent of Happiness given Region.

H_1 (Alternative Hypothesis):

Economic Satisfaction is NOT conditionally independent of Happiness given Region.

Assessment of Odds Ratio Homogeneity:

The odds ratios in Urban (0.22) and Rural (0.14) strata are both less than 1 and in the same direction, indicating that the assumption of homogeneous odds ratios is reasonably satisfied.

```
> urban_or1
[1] 0.2197943
> rural_or1
[1] 0.1359375
```

```
        Mantel-Haenszel chi-squared test with continuity correction

data:  table_cmh1
Mantel-Haenszel X-squared = 158.65, df = 1, p-value < 2.2e-16
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 0.1617502 0.2713174
sample estimates:
common odds ratio
         0.209489
```

- Reject the null hypothesis at 5% significance level, The Mantel–Haenszel chi-squared test indicates a highly significant association between Happiness and Economic Satisfaction after controlling for Region (X^2 = 158.65, df = 1, $p < 2.2e-16$), The common odds ratio of 0.21 (95% CI: 0.16–0.27) shows that individuals in one category of Economic Satisfaction have substantially lower odds of being happy compared to the reference category.

5.2 Happiness, Religion and Gender

H_0 (Null Hypothesis):

Religion is conditionally independent of Happiness given Gender.

H_1 (Alternative Hypothesis):

Religion is NOT conditionally independent of Happiness given Gender.

Assessment of Odds Ratio Homogeneity:

The odds ratios for Males (2.00) and Females (1.998) are almost identical and in the same direction, indicating that the assumption of homogeneous odds ratios across gender strata is well satisfied.

```
> m_or1
[1] 2.001991
> f_or1
[1] 1.997512
```

```
        Mantel-Haenszel chi-squared test with continuity correction

data:  table_cmh2
Mantel-Haenszel X-squared = 31.943, df = 1, p-value = 1.588e-08
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 1.571828 2.544803
sample estimates:
common odds ratio 
         1.999998 
```

- Reject the null hypothesis at 5% significance level, The Mantel–Haenszel chi-squared test shows a highly significant association between Happiness and Religion after controlling for Gender ($X^2 = 31.943$, df = 1, p = 1.59e-08). The common odds ratio is 2.00 (95% CI: 1.57–2.54), indicating that individuals in one category Religion have approximately twice the odds of being happy compared to the reference category.

From the previous we can conclude that the analysis demonstrated that happiness is significantly associated with several demographic and social variables, even after accounting for the influence of a third variable. Three-way contingency tables and mosaic plots showed clear deviations from independence, which were confirmed by chi-square and Mantel–Haenszel tests. Age, education, economic satisfaction, respect for authority, and religion all exhibit strong relationships with happiness within different regions or genders. Economic satisfaction and religion showed the strongest effects, with stable and homogeneous odds ratios across strata. Overall, the findings indicate that happiness is not conditionally independent of the variables studied, and these factors jointly shape patterns of well-being across the population.

Conclusion:

The findings indicate that **subjective well-being (happiness) is not conditionally independent of the variables studied**, confirming that a combination of economic factors, social values, and demographic characteristics jointly shape patterns of well-being across the population.

The key conclusions are:

- **Strongest Predictors (Economic and Religious Values):**
    - **Economic Satisfaction** emerged as the single strongest predictor of happiness. The association was highly significant and robust, remaining stable even when controlled for geographical region. Individuals dissatisfied with their household finances were substantially more likely to report being **NOT happy**.
    - **Religion Importance** was also found to be a strong and statistically significant predictor. Individuals who considered religion **Not Important** were approximately twice as likely to report being **NOT happy** compared to those who viewed it as important. This positive association was consistent across both male and female groups.
    - **Respect for Authority** showed a significant, positive, albeit modest, association with happiness, suggesting that higher commitment to authority is linked to higher self-reported happiness.
- **Weak but Significant Predictors (Demographics):**
    - **Age** and **Education Level** both demonstrated statistically significant associations with happiness; however, the strength of these relationships was quantified as **very weak**. This suggests that while they play a role, their influence is minor compared to economic and religious factors.
    - The association between age and happiness was found to vary by region: happiness peaked among middle-aged adults in urban areas but was highest among older adults (50+) in rural areas.

- **Non-Significant Factors:**
    - The demographic factors of **Gender** and **Region (Urban vs. Rural)**, along with the social value of **Tolerance**, showed **no statistically significant relationship** with overall happiness in the final analysis. This implies that—after controlling other variables where a person lives (urban vs. rural) and their gender does not reliably predict their level of subjective well-being in this US sample.

In summary, the research highlights that in the American context, **financial well-being and core spiritual/social values (religion and respect for authority)** are the primary non-demographic pillars of subjective happiness.

Appendix:

```r
library(readxl)
library(dplyr)

df <- read_excel("USA.xlsx")
View(df)

cat("===== Unique values BEFORE cleaning =====\n")
for(col in names(df)){
  cat("\n---", col, "---\n")
  print(unique(df[[col]]))
}
summary(df)
gtsummary::tbl_summary(df[,1])
gtsummary::tbl_summary(df[,2])
gtsummary::tbl_summary(df[,3])
gtsummary::tbl_summary(df[,4])
gtsummary::tbl_summary(df[,5])
gtsummary::tbl_summary(df[,6])
gtsummary::tbl_summary(df[,7])
gtsummary::tbl_summary(df[,8])
gtsummary::tbl_summary(df[,9])

# 2) Identify numeric and categorical variables
numeric_vars <- names(df)[sapply(df, is.numeric)]
categorical_vars <- setdiff(names(df), numeric_vars)

cat("\nNumeric variables:\n")
print(numeric_vars)

cat("\nCategorical variables:\n")
print(categorical_vars)

cat("\nNumber of numeric variables: ", length(numeric_vars), "\n")
cat("Number of categorical variables: ", length(categorical_vars), "\n\n")

# 3) Replace all negative values with NA
df_clean <- df %>% mutate(across(everything(),
                                 ~ ifelse(.x < 0, NA, .x)))

# 4) Display unique values after removing negative values
cat("===== Unique values AFTER cleaning =====\n")
for(col in names(df_clean)){
  cat("\n---", col, "---\n")
  print(unique(df_clean[[col]]))
}
```

```r
# df_clean is the dataset after removing negative values

df_final <- df_clean %>%
  mutate(
    # Create a new variable Q50_recoded
    Q46 = case_when(
      Q46 %in% 1:2  ~ "HAPPY",
      Q46 %in% 3:4 ~ "NOT happy"
    ),

    # Convert the new variable into a factor
    Q46 = factor(
      Q46,
      levels = c("HAPPY", "NOT happy"),
    )
  ) %>%
  rename(happiness=Q46) %>%

  mutate(Q6 = factor(Q6,
                     levels = c(1,2,3,4),
                     labels = c("Very important", "Rather important","Not very important","Not at all important"))) %>%
  mutate(
    # Recode the four categories into two categories
    Q6 = case_when(
      Q6 %in% c("Very important", "Rather important") ~ "Important",
      Q6 %in% c("Not very important", "Not at all important") ~ "Not Important",
      TRUE ~ NA_character_
    ),

    # Convert the new variable into a factor
    Q6 = factor(
      Q6,
      levels = c("Important", "Not Important")
    )
  )%>%
  rename(religion=Q6) %>%

  # Q12: binary
  mutate(Q12 = factor(Q12,
                      levels = c(1,2),
                      labels = c("Yes", "No"))) %>%
  rename(tolerance=Q12) %>%

  # Q45: ordinal 1-3
  mutate(Q45 = factor(Q45,
                      levels = c(1,2,3),
                      labels = c("Very_interested","Somewhat","Not_interested"),
                      ordered = TRUE)) %>%
  rename(authority_respect=Q45) %>%
```

```r
  # Q50: recode into two categories (Dissatisfied / Satisfied)
  mutate(
    # Create a new variable Q50_recoded
    Q50 = case_when(
      Q50 %in% 1:5  ~ "Dissatisfied",
      Q50 %in% 6:10 ~ "Satisfied"
    ),

    # Convert the new variable into a factor
    Q50 = factor(
      Q50,
      levels = c("Dissatisfied", "Satisfied"),
    )
  ) %>%
  rename(economical_satisfication=Q50) %>%

  # Q260: gender
  mutate(Q260 = factor(Q260,
                       levels = c(1,2),
                       labels = c("M","F"))) %>%
  rename(gender=Q260) %>%

  # X003R: age 1-3 → ordinal
  mutate(X003R2 = factor(X003R2,
                         levels = c(1,2,3),
                         labels = c("16-29 years","30-49 years","50 and more years"
                         ),
                         ordered = TRUE)) %>%
  rename(age=X003R2) %>%

  # H_URBRURAL: urban/rural
  mutate(H_URBRURAL = factor(H_URBRURAL,
                             levels = c(1,2),
                             labels = c("Urban","Rural"))) %>%
  rename(region=H_URBRURAL) %>%

  # Q275R: education (1-3) → ordinal
  mutate(Q275R = factor(Q275R,
                        levels = c(1,2,3),
                        labels = c("Lower","Middle","Higher"),
                        ordered = TRUE)) %>%
  rename(education=Q275R)

df_final <- df_final %>%
  tidyr::drop_na()

str(df_final)
summary(df_final)
```

```r
###########Describing categorical variables###################-----------------------
######### 2*2 contigency table for Religion and  happiness
table2 <-gtsummary::tbl_cross(row=2L,col=1L,data=df_final)
# the joint probabilities
gtsummary::tbl_cross(row=2L,col=1L,data=df_final, percent = "cell")
# the conditional probablities by Religion
gtsummary::tbl_cross(row=2L,col=1L,data=df_final, percent = "row")
gtsummary::tbl_cross(row=2L,col=1L,data=df_final, percent = "column")
#plot for happiness and Religion
library(ggplot2)
ggplot(df_final, aes(x = religion, fill = happiness)) +
  geom_bar(position = "dodge") +
  labs(title = "Happiness distribution by religion importance") +
  theme_minimal()
library(vcd)
mosaic(~ religion + happiness, data = df_final,
       shade = TRUE,
       legend = TRUE)
######### 2*2 contigency table for tolerance and  happiness
gtsummary::tbl_cross(row=3L,col=1,data=df_final)
# the joint probabilities
gtsummary::tbl_cross(row=3L,col=1,data=df_final,percent = 'cell')
# the conditional probablities by tolerance
gtsummary::tbl_cross(row=3L,col=1,data=df_final,percent = 'column')
gtsummary::tbl_cross(row=3L,col=1,data=df_final,percent = 'row')
#plot for happiness and tolerance
library(ggplot2)
ggplot(df_final, aes(x = tolerance, fill = happiness)) +
  geom_bar(position = "dodge") +
  labs(title = "Happiness distribution by tolerance") +
  theme_minimal()
library(vcd)
mosaic(~ tolerance + happiness, data = df_final,
       shade = TRUE,
       legend = TRUE)

######### 2*2 contigency table for authority respect and  acceptance_of_immigrates
gtsummary::tbl_cross(row=4L,col=1,data=df_final)
# the joint probabilities
gtsummary::tbl_cross(row=4L,col=1,data=df_final,percent = 'cell')
# the conditional probablities by authority respect
gtsummary::tbl_cross(row=4L,col=1,data=df_final,percent = 'column')
gtsummary::tbl_cross(row=4L,col=1,data=df_final,percent = 'row')
```

```
#plot for happiness and age
library(ggplot2)
ggplot(df_final, aes(x = authority_respect, fill = happiness)) +
  geom_bar(position = "dodge") +
  labs(title = "Happiness distribution by authority_respect") +
  theme_minimal()
library(vcd)
mosaic(~ authority_respect + happiness, data = df_final,
       shade = TRUE,
       legend = TRUE)
######### 2*2 contigency table for economical satisfication and  happiness
gtsummary::tbl_cross(row=5L,col=1,data=df_final)
# the joint probabilities
gtsummary::tbl_cross(row=5L,col=1,data=df_final,percent = 'cell')
# the conditional probablities by economical satisfication
gtsummary::tbl_cross(row=5L,col=1,data=df_final,percent = 'column')
gtsummary::tbl_cross(row=5L,col=1,data=df_final,percent = 'row')
#plot for happiness and economical satisfication
library(ggplot2)
ggplot(df_final, aes(x = economical_satisfication, fill = happiness)) +
  geom_bar(position = "dodge") +
  labs(title = "Happiness distribution by economical satisfication") +
  theme_minimal()
library(vcd)
mosaic(~ economical_satisfication + happiness, data = df_final,
       shade = TRUE,
       legend = TRUE)

######### 2*2 contigency table for gender and  happiness
gtsummary::tbl_cross(row=6L,col=1,data=df_final)
# the joint probabilities
gtsummary::tbl_cross(row=6L,col=1,data=df_final,percent = 'cell')
# the conditional probablities by gender
gtsummary::tbl_cross(row=6L,col=1,data=df_final,percent = 'column')
gtsummary::tbl_cross(row=6L,col=1,data=df_final,percent = 'row')
#plot for happiness and gender
library(ggplot2)
ggplot(df_final, aes(x = gender, fill = happiness)) +
  geom_bar(position = "dodge") +
  labs(title = "Happiness distribution by gender") +
  theme_minimal()
library(vcd)
mosaic(~ gender + happiness, data = df_final,
       shade = TRUE,
       legend = TRUE)
```

```r
########2*2 contigency table for AGE and  Happiness
table1 <- gtsummary::tbl_cross(row=7L,col=1L,data=df_final)
# the joint probabilities
gtsummary::tbl_cross(row=7L,col=1L,data=df_final, percent = "cell")
# the conditional problities by age
gtsummary::tbl_cross(row=7L,col=1L,data=df_final, percent = "row")
gtsummary::tbl_cross(row=7L,col=1L,data=df_final, percent = "column")
#plot for happiness and age
library(ggplot2)
ggplot(df_final, aes(x = age, fill = happiness)) +
  geom_bar(position = "dodge") +
  labs(title = "Happiness distribution by age") +
  theme_minimal()
library(vcd)
mosaic(~ age + happiness, data = df_final,
       shade = TRUE,
       legend = TRUE)

######### 2*2 contigency table for region and  happiness
gtsummary::tbl_cross(row=8L,col=1,data=df_final)
# the joint probabilities
gtsummary::tbl_cross(row=8L,col=1,data=df_final,percent = 'cell')
# the conditional problities by education
gtsummary::tbl_cross(row=8L,col=1,data=df_final,percent = 'column')
gtsummary::tbl_cross(row=8L,col=1,data=df_final,percent = 'row')
#plot for happiness and region
library(ggplot2)
ggplot(df_final, aes(x = region, fill = happiness)) +
  geom_bar(position = "dodge") +
  labs(title = "Happiness distribution by region") +
  theme_minimal()
library(vcd)
mosaic(~ region + happiness, data = df_final,
       shade = TRUE,
       legend = TRUE)
```

```
######### 2*2 contigency table for education and  happiness
gtsummary::tbl_cross(row=9L,col=1,data=df_final)
# the joint probabilities
gtsummary::tbl_cross(row=9L,col=1,data=df_final,percent = 'cell')
# the conditional probablities by education
gtsummary::tbl_cross(row=9L,col=1,data=df_final,percent = 'column')
gtsummary::tbl_cross(row=9L,col=1,data=df_final,percent = 'row')
#plot for happiness and education
library(ggplot2)
ggplot(df_final, aes(x = education, fill = happiness)) +
  geom_bar(position = "dodge") +
  labs(title = "Happiness distribution by educational level") +
  theme_minimal()
library(vcd)
mosaic(~ education + happiness, data = df_final,
       shade = TRUE,
       legend = TRUE)


##################################################################################
library(epitools)
table1 <- table(df_final$region,df_final$happiness)
table2 <- table(df_final$religion,df_final$happiness)
table3 <-table(df_final$tolerance,df_final$happiness)
table4 <-table(df_final$economical_satisfication,df_final$happiness)
table5 <- table(df_final$gender,df_final$happiness)
table6 <- table(df_final$education,df_final$happiness)
table7 <- table(df_final$age,df_final$happiness)
table8 <- table(df_final$authority_respect,df_final$happiness)
#odds ratio for the binary varibles
epitools::oddsratio(table1)
epitools::oddsratio(table2)
epitools::oddsratio(table3)
epitools::oddsratio(table4)
epitools::oddsratio(table5)

#test of independece
chisq.test(table1,correct=FALSE)
chisq.test(table2,correct=FALSE)
chisq.test(table3,correct=FALSE)
chisq.test(table4,correct=FALSE)
chisq.test(table5,correct=FALSE)
chisq.test(table6,correct=FALSE)
chisq.test(table7,correct=FALSE)
chisq.test(table8,correct=FALSE)
```

```
#measure of assoication
library(DescTools)
DescTools::Assocs(table2, conf.level = 0.95, verbose = NULL)
DescTools::Assocs(table4, conf.level = 0.95, verbose = NULL)
DescTools::Assocs(table6, conf.level = 0.95, verbose = NULL)
DescTools::Assocs(table7, conf.level = 0.95, verbose = NULL)
DescTools::Assocs(table8, conf.level = 0.95, verbose = NULL)




#3_way contiency table

#age, happiness and region
tab1 <- xtabs( ~ happiness +age + region , data = df_final)
View(tab1)
library(janitor)
table_3way1 <- df_final %>%
  tabyl( happiness, age,region) %>%
  adorn_totals("col") %>%
  adorn_totals("row") %>%
  adorn_title("combined")

# happiness , economical_satisfication and region

tab2 <- xtabs( ~ happiness + economical_satisfication + region , data = df_final)
View(tab2)
table_3way2 <- df_final %>%
  tabyl(happiness,economical_satisfication , region) %>%
  adorn_totals("col") %>%
  adorn_totals("row") %>%
  adorn_title("combined")
# Happiness , Education and Region
tab3 <- xtabs( ~ happiness + education + region , data = df_final)
View(tab3)
table_3way3 <- df_final %>%
  tabyl(happiness, education, region) %>%
  adorn_totals("col") %>%
  adorn_totals("row") %>%
  adorn_title("combined")
#Happiness , Authority_respect and Gender

tab4 <- xtabs( ~ happiness + authority_respect + gender , data = df_final)
View(tab4)
table_3way4 <- df_final %>%
  tabyl(happiness,authority_respect,gender ) %>%
  adorn_totals("col") %>%
  adorn_totals("row") %>%
  adorn_title("combined")
```

```
# Happiness , Religion and Gender
tab5 <- xtabs( ~ happiness + religion + gender , data = df_final)
View(tab5)
table_3way5 <- df_final %>%
  tabyl(happiness,religion,gender ) %>%
  adorn_totals("col") %>%
  adorn_totals("row") %>%
  adorn_title("combined")
# mosaic plots
# Load necessary library
library(vcd)
mosaic(tab1, shade=TRUE, legend=TRUE,
       ,main ="Mosaic Plot: Happiness ,Age and Region ")
mosaic(tab5, shade=TRUE, legend=TRUE,
       ,main ="Mosaic Plot:Happiness , Religion and Gender ")
#assoication in 3_way
library(janitor)
table_3way11=df_final %>%
  tabyl(happiness,age,region)
table_3way11$Urban
table_3way11$Rural

x=chisq.test(table_3way11$Urban)
y=chisq.test(table_3way11$Rural)


table_3way33=df_final %>%
  tabyl(happiness,education,region)
table_3way33$Urban
table_3way33$Rural

a=chisq.test(table_3way33$Urban)
b=chisq.test(table_3way33$Urban)

table_3way44=df_final %>%
  tabyl(happiness,authority_respect,gender)
table_3way44$M
table_3way44$F

s=chisq.test(table_3way44$M)
m=chisq.test(table_3way44$F)

# checking CMH assumptions
# Load packages
library(dplyr)
library(DescTools)
#happiness,economical_satisfication,region
# 2) Extract each region's 2x2 numeric table
urban_tab1  <- tab2[, , "Urban"]
rural_tab1  <- tab2[, , "Rural"]
```

```
# association using relative risk
# RR for urban
rr_urban <- riskratio(urban_tab1, rev = "columns")
rr_male

# RR for rural
rr_rural <- riskratio(rural_tab1, rev = "columns")
rr_rural

# 3) Compute odds ratios for each region
urban_or1  <- DescTools::OddsRatio(urban_tab1)
rural_or1  <- DescTools::OddsRatio(rural_tab1)
# 4) Print results
urban_or1
rural_or1
# use the CMH test
table_cmh1=xtabs( ~ happiness+economical_satisfication+region  , data =df_final)

mantelhaen.test(table_cmh1)
#  Happiness , Religion and Gender

# 2) Extract each region's 2x2 numeric table
m_tab1  <- tab5[, , "M"]
f_tab1  <- tab5[, , "F"]
# association using relative risk
rr_male <- riskratio(m_tab1, rev = "columns")
rr_male

# RR for females
rr_female <- riskratio(f_tab1, rev = "columns")
rr_female

# 3) Compute odds ratios for each region
m_or1  <- DescTools::OddsRatio(m_tab1)
f_or1  <- DescTools::OddsRatio(f_tab1)

# 4) Print results
m_or1
f_or1
# use the CMH test
table_cmh2=xtabs( ~ happiness+religion+gender  , data =df_final)

mantelhaen.test(table_cmh2)
```